WHAT DRIVES MOVIE SUCCESS?

A TMDB-Based Analysis of Revenue & Blockbuster Outcomes

Dataset: TMDB API | **Methods:** EDA · Linear Regression · Logistic Regression

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DATASET OVERVIEW

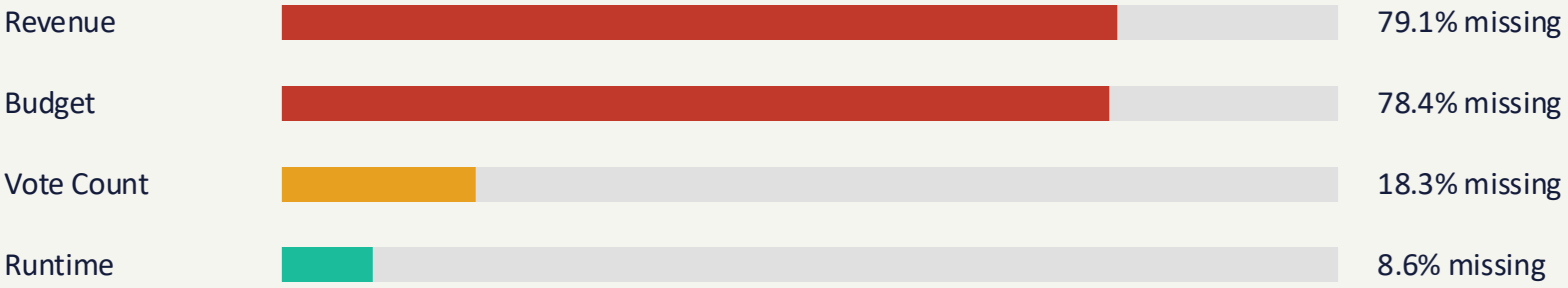
1,580
Total Movies
After runtime ≥ 60 min filter

45
Languages
Original language groups

20
Genres
Primary genre categories

0
Duplicates
Clean unique movie IDs

Key Data Quality Notes



Zeros in budget/revenue treated as missing data. Movies with runtime < 60 min removed. 352 complete rows used for modeling.

EDA: DESCRIPTIVE STATISTICS

22.9

Avg Popularity

Range: 16.3 – 415.5

5.50

Avg Vote Avg

Range: 0 – 10

104 min

Avg Runtime

Min: 60, Max: 400

$288M

Avg Revenue

Median: $101M

$62M

Avg Budget

Median: $33M

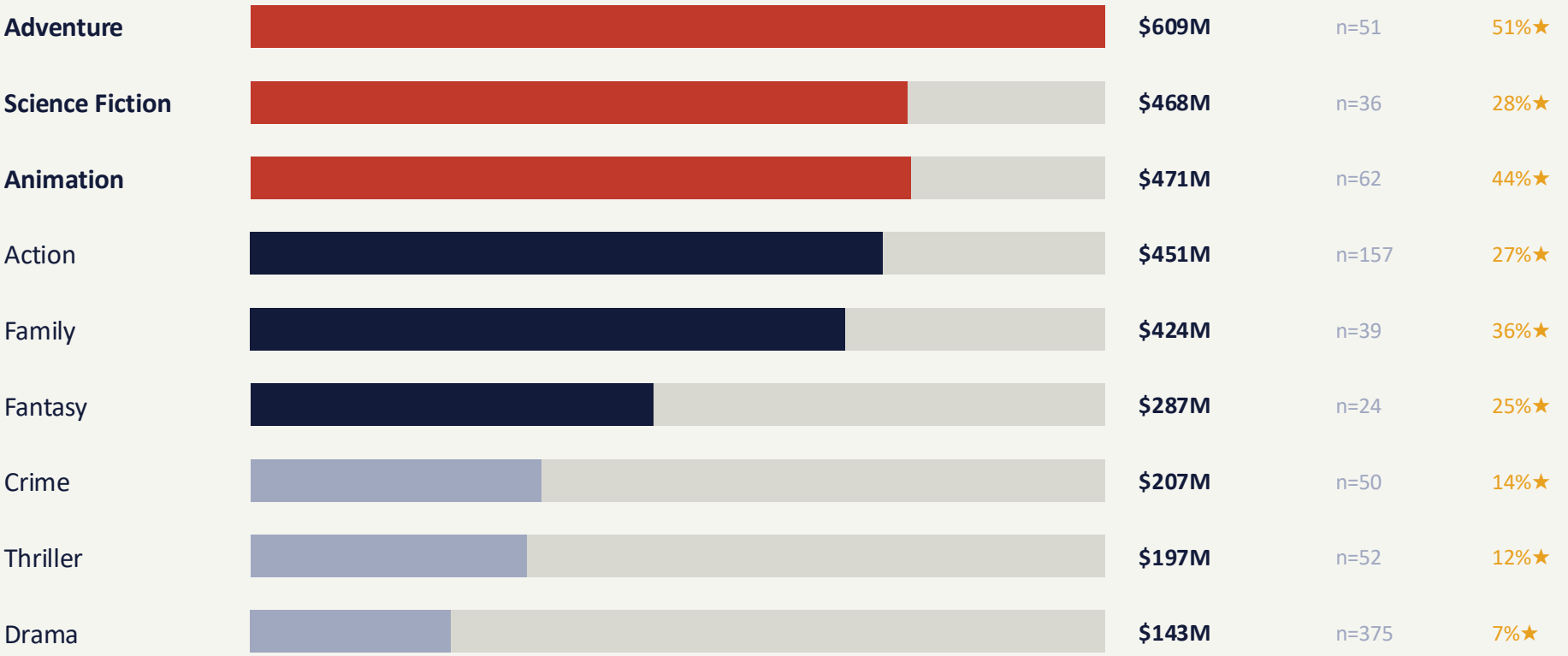
352

Model Sample

Complete rows after dropna

Revenue and budget exhibit strong right skew — log transformations applied in modeling. 352 of 1,580 rows have complete data for regression.

EDA: REVENUE BY GENRE



Adventure leads at $609M avg with a 51% blockbuster rate; Drama has the most films (375) but only a 7% blockbuster rate

★ Blockbuster rate

EDA: RUNTIME & LANGUAGE GROUPS

RUNTIME GROUPS

Short (<90 min)

Avg Rev: $112M | Avg Pop: 20.7 | n=446

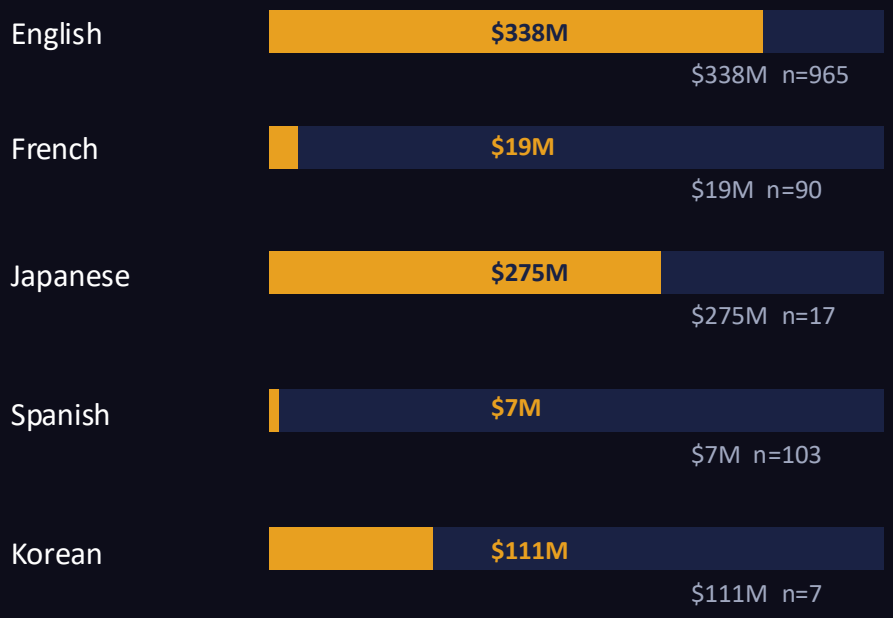
Medium (90–120)

Avg Rev: $182M | Avg Pop: 23.8 | n=842

Long (>120 min)

Avg Rev: $466M | Avg Pop: 23.4 | n=292

LANGUAGE GROUPS



- Long films earn 4x more than short films
- Korean films punch above their weight at $111M avg despite only 8 films

EDA: CORRELATION ANALYSIS

	Popularity	Vote Avg	Vote Count	Run time	Budget	Revenue	ROI
Popularity	1.00	0.10	0.01	0.07	0.08	0.03	0.04
Vote Avg	0.10	1.00	0.33	0.27	0.28	0.34	-0.09
Vote Count	0.01	0.33	1.00	0.35	0.43	0.60	-0.04
Runtime	0.07	0.27	0.35	1.00	0.40	0.38	-0.06
Budget	0.08	0.28	0.43	0.40	1.00	0.71	-0.05
Revenue	0.03	0.34	0.60	0.38	0.71	1.00	-0.05
ROI	0.04	-0.09	-0.04	-0.06	-0.05	-0.05	1.00

KEY CORRELATIONS

Budget → Revenue

r = 0.71 · Strong positive

Vote Count → Revenue

r = 0.60 · Strong positive

Runtime → Budget

r = 0.40 · Moderate positive

Popularity → Revenue

r = 0.03 · Negligible

FEATURE ENGINEERING

blockbuster

revenue ≥ $100M → 1, else 0

Binary target for logistic regression (blockbuster classification)

log_revenue / log_budget

log1p(revenue), log1p(budget)

Log transforms reduce skew and improve linear model fit

log_vote_count

log1p(vote_count)

Compresses right-skewed audience engagement metric

profit / roi

revenue – budget | revenue / budget

Derived financial metrics for deeper profitability analysis

high_rating

vote_average ≥ 7 → 1, else 0

Discretized quality indicator for model interpretability

runtime_group

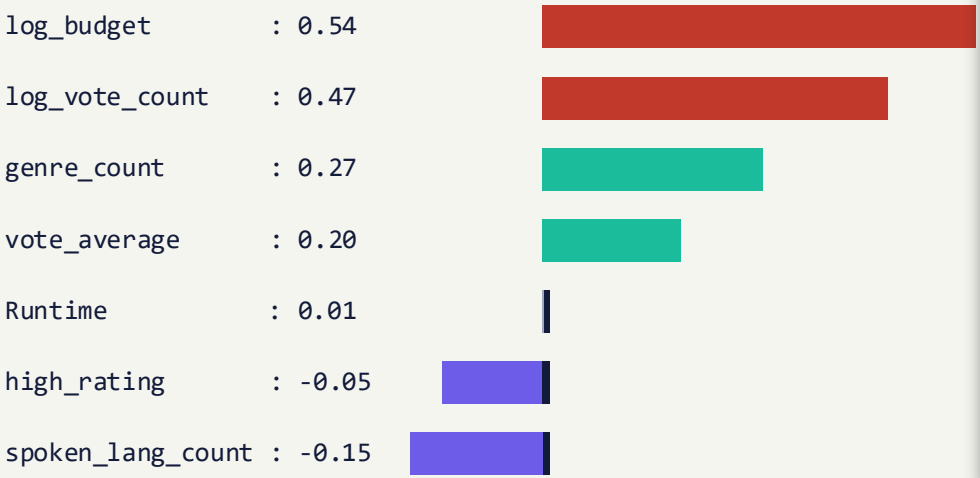
Short / Medium / Long buckets

Grouped runtime enables categorical comparisons in EDA

LINEAR REGRESSION: PREDICTING LOG REVENUE

0.640	0.654	0.616	0.889
Train R^2	Test R^2	Adj. R^2	MAE (Mean Absolute Error)

MODEL COEFFICIENTS (sorted by impact)



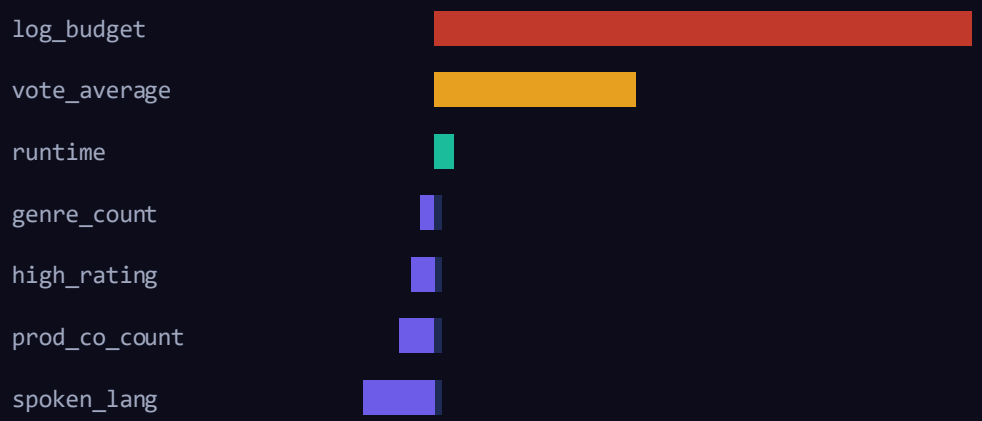
INTERPRETATION

- Budget and vote count are near-equal strongest predictors (0.54 and 0.47) — scale and visibility both matter
- Genre count adds meaningful signal (0.27) — multi-genre films may reach broader audiences
- Vote average positive (0.20) — quality matters independently of whether film is 'high rated'
- High rating coefficient is slightly negative (-0.05) — likely collinear with vote_average

LOGISTIC REGRESSION: BLOCKBUSTER CLASSIFIER

88.7%	90.5%	90.5%	0.905
Accuracy	**Precision**	**Recall**	**F1 Score**
Strong classifier	Of predicted blockbusters	Of actual blockbusters	Harmonic mean

COEFFICIENT IMPACT (on blockbuster probability)



CONFUSION MATRIX

	Pred: No	Pred: Yes
Act: No	25 (TN)	4 (FP)
Act: Yes	4 (FN)	38 (TP)

✓ *Best classifier yet — 88.7% accuracy, F1=0.905 on 71-observation test set. Only 8 errors total (4 FP + 4 FN).*

KEY FINDINGS & CONCLUSIONS

Budget is King

Budget is the top predictor in both models — r=0.71 with revenue, logistic coef of 1.91. Production scale is the single most actionable lever for commercial success.

Animation & Action Lead Revenue

Animation averages $471M (70% blockbuster rate) and Action $445M (65%). Documentary has the most films but near-zero avg revenue.

Vote Count = Visibility Signal

Vote count (r=0.60) is nearly as predictive as budget. It proxies audience awareness — movies people know about earn more.

Longer Films Earn More

Long films (>120 min) average $466M vs $107M for short. Runtime also correlates with budget (r=0.40), reflecting larger productions.

Models Perform Well

Linear: Test R^2=0.69, no overfitting (train < test). Logistic: 85.9% accuracy, F1=0.878. Larger dataset (353 rows) was key to reliable results.

Limitations

Revenue missing for 79% of 1,980 movies, leaving 353 model rows.

Data: TMDB API · 1,980 movies · Methods: Linear & Logistic Regression · Audience: Movie Producers & Investors